SILTHAUL · TRACEABILITY MATRIX

SiltHaul traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/silthaul		
Source	docs/03b-traceability.md @ 9347c2b		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-11-03	Amish Chadha	Generated by .kit/trace.py
0.2	2026-11-03	Amish Chadha	Generated by .kit/trace.py
0.1	2026-11-03	Amish Chadha	Generated by .kit/trace.py

SiltHaul traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 13 user requirements, 18 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001, UR-002	PR-001	Mud moved per hour by a crew of four	SLH-CAL-001, SLH-BLD-001, SLH-STM-001	Side-by-side timed trial on a test bed of wet silt (TRL 4)	Met on paper, at risk: 0.51 against 0.90 m3/h (ratio 0.57) with a 1.5 min fill; 0.497 if the loaders fill at a shoveller's 10 L/min (SLH-CAL-001 D6); loaders push mud in over the lip; nobody lifts or carries it
UR-003	PR-002	Crank handle force at full box	SLH-CAL-001, SLH-BLD-001, SLH-STM-001	Spring balance at the handle during trial (TRL 4)	Met on paper: 63 N each at 1,000 N (126 N for one person); 61 N each at the 977 N stroke estimate
UR-002	PR-003	Scraper box capacity	SLH-CAL-001, SLH-BLD-001	Measured fill on test bed (TRL 4)	Met by design: 40.3 L in each of the two boxes at a 245 mm fill
UR-004	PR-004	Fits through a doorway	SLH-CAL-001, SLH-BLD-001	Fit test (TRL 4)	Met by design: each box 258 mm over the shackles; the boxes pass 42 mm apart and cross the ramp 26 mm inside its cheeks; ramp 646 mm
UR-005	PR-005	Working length	SLH-CAL-001, SLH-BLD-001	Trial at full length (TRL 4)	Met by design: 12 m of travel on each drum half; capstan at least 6 m from the door
UR-006	PR-006	Rope and anchor strength	SLH-CAL-001, SLH-BLD-001, SLH-STM-001, SLH-OPT-001	CalRig proof-load test of each anchor type (TRL 4)	At risk: rope 18 times working, 5.4 at the limit; chain 4.93 at the 3,000 N limit (5.03 at the 2,941 N the pin can release); fix proposed to Amish (SLH-STM-001 S5)
UR-006	PR-007	Tail anchor independent of walls	SLH-CAL-001, SLH-BLD-001, SLH-OPT-001	Pull test on the anchored tail block (TRL 4)	Met on paper on a sound slab: 0.45 of recommended anchor loads
UR-007	PR-008	Setup time	SLH-CAL-001, SLH-BLD-001, SLH-OPT-001	Timed trial (TRL 4)	To verify at TRL 4: about 20 min estimated, at the limit (bolting the frame and shackling two boxes add 4 min)
UR-008	PR-009	Portability	SLH-CAL-001, SLH-BLD-001, SLH-STM-001	Weigh and pack test (TRL 4)	Met on paper, as restated by Amish (2A, round 2): heaviest lift 22.2 kg (drum with bearings); flat-pack frame, largest piece 890 x 814 x 83 mm; kit about 407 L packed against a 594 L small car boot
UR-009	PR-010	Cost	SLH-CAL-001	Costed bill of materials	Met on paper, within the target: USD 1,055.60 (USD 944.40 under)
UR-006	PR-011	Overload limit	SLH-CAL-001, SLH-BLD-001,	Shear pin release test on the capstan (TRL 4)	Met on paper: shear pin releases at 2,451 N (1,961 to 2,941 N)

User requirement	Product requirement	Title	Evidence	Verification	Status
			SLH-STM-001, SLH-OPT-001		
UR-006	PR-012	Holds when let go	SLH-CAL-001, SLH-BLD-001, SLH-OPT-001	Release test at working pull (TRL 4)	Met by design: two ratchet wheels and pawls on the drum shaft
UR-010	PR-013	Made from stock	SLH-CAL-001	Analysis of the BOM and the 245-part model (TRL 3); prototype build (TRL 4)	Met by design: 40 x 40 x 2 SHS, 219.1 x 3 tube, sheet and bar; UCP bearings, 08B-1 chain and sprockets, rope and fasteners bought
UR-011	PR-014	Survives mud and wet	SLH-CAL-001	Wash-down and inspection after a trial (TRL 4)	To verify at TRL 4: coatings and sealed UCP bearings in the BOM; coating life not assessed
UR-012	PR-015	Crank crew clear of the mud	SLH-CAL-001	Layout drawing SLH-DWG-002; timed trial (TRL 4)	Met by design: capstan at least 6 m from the ramp roller (fleet angle 1.41 degrees); loaders push mud in at floor level
UR-006	PR-016	Moving parts guarded	SLH-CAL-001	Inspection of the prototype (TRL 4)	Met by design: closed 1.5 mm sheet chain guard (BOM 9) held by two M6 screws
UR-003	PR-017	Tipping a full box	SLH-CAL-001, SLH-STM-001	Spring balance at the bar (TRL 4)	Met on paper: 237 N to start a full box, 119 N each with two people (SLH-CAL-001 K3)
UR-013	PR-018	No added water	SLH-CAL-001	Analysis (TRL 3)	Met by design for option A (no flush water); option C would deliver 2.8 m3 per m3 (SLH-OPT-001)

Gaps

None.